

Sri Lanka Institute of Information Technology



SE3090 – Software Engineering Frameworks
Year 3, Semester 1 - 2026

Practical Answer Submission

Practical Sheet No: 01

Student ID	IT24102798
Student Name	Sooriyabandara U.R.G.W.K.
Campus/ Center name	Malabe
Specialization	Artificial Intelligence
Batch No.	01.01

Task 01

1.

Technology	Framework/Library	Justification (Who controls the flow?)
lodash	Library	The application calls lodash functions whenever required, so the application controls the execution flow.
ASP.NET Core	Framework	ASP.NET Core manages the application lifecycle and calls controllers and middleware at the appropriate stages.
Axios	Library	Axios is called by the application to send HTTP requests, meaning the application controls when it is executed.
Flutter	Framework	Flutter controls the application lifecycle, rendering process, and event handling while invoking developer-defined widgets.
jQuery	Library	jQuery functions are called by the application to manipulate the DOM and handle events, so it does not control the application flow.
Angular	Framework	Angular manages the application's lifecycle, routing, and dependency injection while calling application components as needed.
Express.js	Framework	Express.js manages the request-response lifecycle and invokes middleware and route handlers when requests are received.
NumPy	Library	NumPy provides functions for numerical computations that are called by the application whenever needed.
React	Library	React is used to render and update the user interface, while the overall application remains responsible for controlling execution.
Spring Boot	Framework	Spring Boot manages application startup, dependency injection, and request handling while invoking application code.

2. *(Done)*

3. *(Done)*

4. Reason for React to consider inside “Grey Zone”,

- Because it provides framework-like features such as component lifecycle management and virtual DOM rendering. However, it is classified as a library since it focuses on building the user interface rather than controlling the entire application lifecycle. Therefore, the application remains responsible for the overall execution flow, making library the most appropriate classification.

Task 02

1. **Chosen Framework Family** – Frontend (Web UI)
2. *(Done)*
- 3.

Criterion	Framework A: React	Framework B: Angular	Framework C: Vue
Type	Library	Framework	Framework
Language	JavaScript / TypeScript	TypeScript	JavaScript / TypeScript
Best Fit / Strengths	Flexible UI development, reusable components, and a large ecosystem.	Suitable for large-scale enterprise applications with built-in features.	Easy to learn, lightweight, and ideal for small to medium-sized projects.
Used By (Industry)	Meta, Netflix, Airbnb	Google, Microsoft, Deutsche Bank	Alibaba, Xiaomi, GitLab
Main Trade-off / Cost	Requires additional libraries for routing and state management.	Has a steeper learning curve and can be more complex for beginners.	Smaller ecosystem and lower enterprise adoption compared to React and Angular.

4. Short Verdict,
 - For a three-developer student startup building its first product within four months, React would be the most suitable choice. It provides flexibility, a large community, and a wide range of third-party libraries that help speed up development. React also has a gentle learning curve compared to Angular while offering a more mature ecosystem than Vue. These advantages make it a practical option for developing a scalable product within a limited timeframe.

Task 03

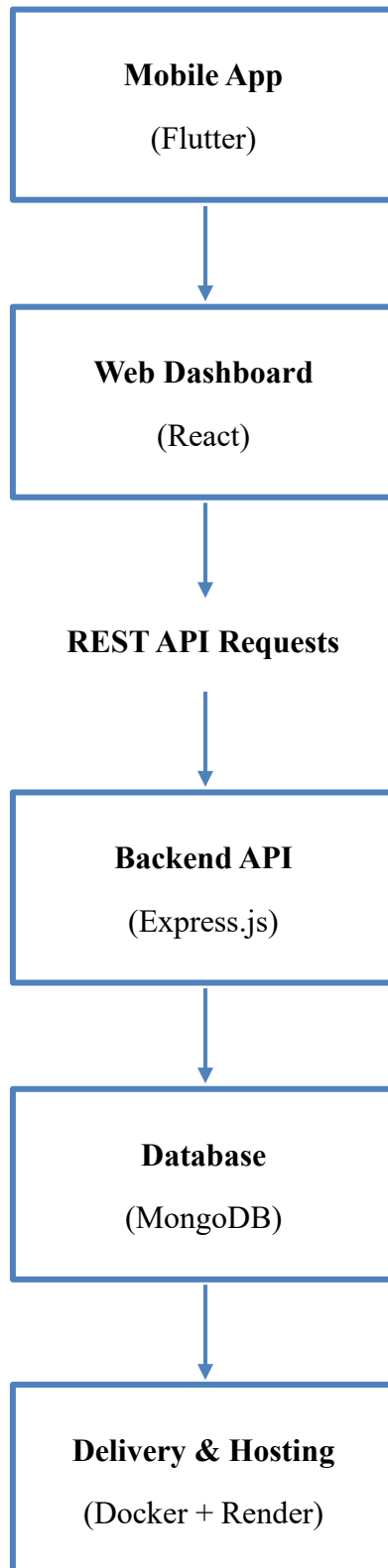
Part A – Weighted Decision Matrix

Criterion	Weight %	Flutter	React Native	Native (SwiftUI / Jetpack Compose)
Team familiarity	25%	4	4	2
Time to first working version	25%	5	4	2
Hiring pool (if team grows)	15%	4	5	3
Ecosystem for our features	20%	5	4	3
Long-term maintainability	15%	5	4	3
Weighted total	100%	4.6	4.15	2.5

Part B – GreenCart Technology Stack Plan

Layer	Your Choice	Justification (Cite One Criterion)
Mobile App	Flutter	Project Fit – A single codebase supports both Android and iOS, reducing development time for a three-developer team.
Web Dashboard	React	Community & Ecosystem – React provides a mature ecosystem with many libraries for building responsive dashboards.
Backend API	Express.js	Team Skills – Express.js is lightweight and enables rapid API development using JavaScript across the stack.
Database	MongoDB	Scalability – MongoDB can efficiently manage growing amounts of application data as the system expands.
Delivery & Hosting	Docker + Render	Cost – Docker simplifies deployment while Render provides an affordable hosting solution suitable for small projects.

Part C - GreenCart Technology Architecture Sketch



Task 04

Part A — Select and justify a stack

1. Chosen Scenario,
 - **Option C** – A clothing brand wants a mobile-first e-commerce app that must scale sharply during seasonal sales and promotions.
- Two Most Important Non-Functional Requirements,
 - **Scalability** – The system must manage a large number of users during seasonal sales without performance issues.
 - **Performance** – The application should provide fast response times to ensure a smooth shopping experience.
2. Chosen Frameworks for,
 - Frontend** – Flutter
 - Backend** – Spring Boot
 - Database** – PostgreSQL
 - Hosting** – AWS Cloud
3. Top 3 Selection Criteria,
 - **Scalability** – The system should support increased traffic during sales and promotions without affecting performance.
 - **Performance** – Fast processing and response times improve the user experience and reduce delays during high demand.
 - **Community & Ecosystem** – Well-established technologies provide reliable documentation, tools, and long-term support for development and maintenance.

Part B — Emerging-trends reflection

Trend	Apply? (Yes/No)	Where / Why	One Risk or Cost
Microservices	Yes	Separate services such as orders, payments, and inventory to improve scalability.	More complex deployment and service management.
Serverless Functions	Yes	Use for tasks such as image processing and sending notifications during peak periods.	Higher costs with frequent execution and possible vendor lock-in.
Cloud-native & DevOps (CI/CD)	Yes	Automate testing, deployment, and scaling to support frequent updates.	Requires additional setup and DevOps knowledge.
AI-assisted / Agentic Development	Yes	Assist with coding, testing, and documentation to improve development productivity.	Generated code must be reviewed to ensure accuracy and security.

Quick Test

Multiple Choice (Choose One)

- Q1. (c) who controls the flow of execution
- Q2. (b) Inversion of Control
- Q3. (b) Flutter (cross-platform)
- Q4. (c) has the highest weighted score against your project's criteria
- Q5. (b) review, test and take responsibility for the output

Short Answer (1–2 sentences each)

Q6.

- **One Benefit** – A large framework provides built-in features that can speed up development and improve project structure.
- **One Cost** – It has a steeper learning curve and may add unnecessary complexity for a small project.

Q7.

- 1st Stage – Requirements
- 2nd Stage – Constraints
- 3rd Stage – Shortlist
- 4th Stage – Prototype
- 5th Stage – Decide & Record

Scenario-based

Q8.

- No. Axios is a library because the application calls its functions whenever HTTP requests are needed. Since the application controls the flow of execution rather than axios, it does not meet the definition of a framework.